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1. Executive Summary

A South African medical aid scheme wanted help from machine learning to speed up processing claims for dread disease (cancer) benefits. The goal of the project is to create a classification model that will allow them to easily identify patients who may be eligible for assistance by predicting if they have breast cancer.

To support this work, the researchers used the breast cancer diagnostic data set from the University of Wisconsin. The diagnosis variable has two values; the “M” for malignant (cancerous) and “B” for benign (non-cancerous). Therefore, solving this problem is a standard binary classification problem. Several models were developed and evaluated. The Random Forest Classifier demonstrated the best performance, achieving an almost perfect test set ROC-AUC of 0.996 and an accuracy of 0.956. The use of this model indicates that the Random Forest Classifier has successfully learned complicated, non-linear relationships in the data and will provide significant assistance in supporting physicians with this diagnosis.

Summary of Findings:

- Most accurate classifier, across all classes, on each performance measure, of Random Forest Classifier is good performance in identifying cancerous tumours (class = Malignant).
- Logistic Regression model performed well, in addition, was beneficial in that it is interpretable, therefore, feature importance can be understood (i.e. worst and mean concave points being strong indicators of being Malignant).
- Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) were ineffective , particularly on the Malignant class due to the class imbalance; therefore have less than expected recalls for the class = Malignant.
- Overall, the results for the Random Forest and Logistic Regression models are very promising and indicate that with some more development of the models and additional data, the resultant model could be deployed.

2. Dataset Selection and Justification

Dataset Source: Breast Cancer Wisconsin (Diagnostic) Data Set

Justification for Choice:

Criterion	Assessment
Relevance	Directly addresses client needs regarding predicting cancer diagnoses (malignant vs. benign).
Quality	Quality – Dataset has 569 samples, 30 numerical features extracted from digitized cell nuclei images with no missing values.
Suitability	All features are numerical, continuous, and require very little preprocessing prior to use with classification algorithms.

Benchmark	A commonly used dataset in medical machine learning literature and has benchmarked previously allowing researchers to compare their performance.
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3. Algorithm Selection Justification

Four classification algorithms were selected for comparison:

Algorithm	Justification
Logistic Regression	A linear model that is simple and easy to interpret, with coefficients representing the weight of each feature in determining the output variable. A Logistic Regression model serves as a performance baseline.
K-Nearest Neighbors (KNN)	A non-parametric, distance-based approach that can create very complex decision boundaries based on the data available.
Support Vector Machine (SVM)	An extremely capable classifier that works well in a high-dimensional space; uses an RBF kernel to model non-linear relationships.
Random Forest	An ensemble method that is robust to outliers and different scales and can create non-linear models as well; provides importance scores for each feature in the model.

4. Data Cleaning and Exploratory Data Analysis

4.1. Data Overview

Total Number of Records: 569

Total Number of Features: 33

Target Variable: type of diagnosis (B= Benign; M= Malignant)

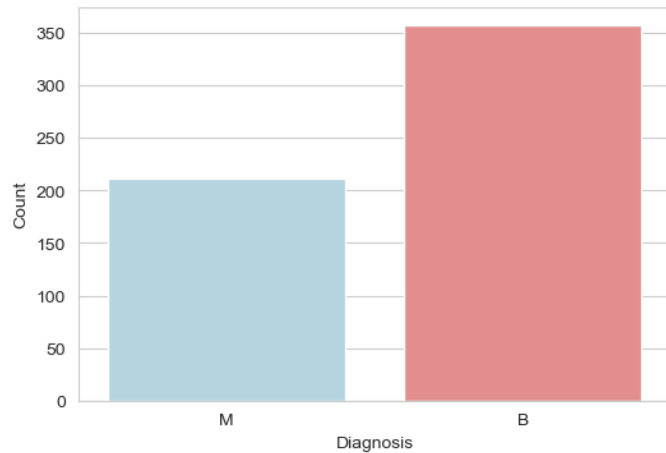
Missing Data: No missing data found

Class Distribution: Benign (357 records), Malignant (212 records)

Important Finding(s): There is an imbalance between the number of cases of malignant and benign diagnoses within this dataset; therefore, the model was trained with `class_weight = 'balanced'` to account for the imbalance (37.3%).

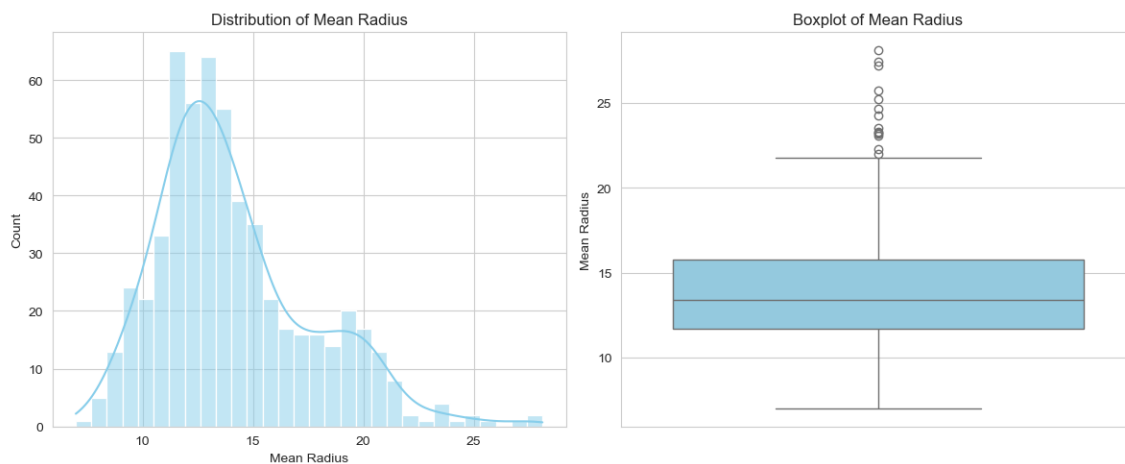
4.2. Visualizations

Figure 1: Distribution of Diagnosis (Malignant vs. Benign)



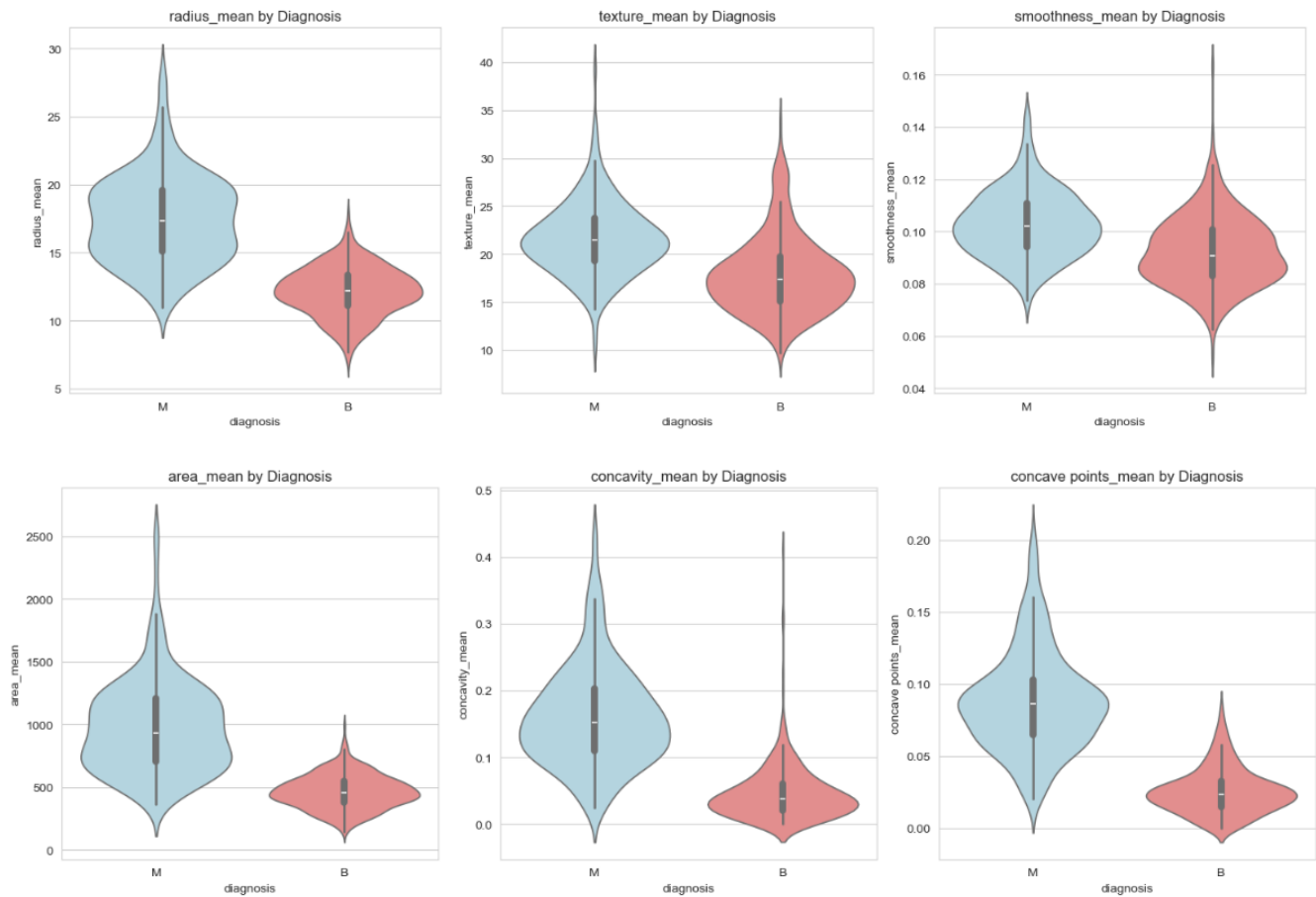
Interpretation: There appears to be some slight class imbalance in this dataset, which is expected and common for datasets used for analyzing medical diagnoses.

FIGURE 2: Distribution of Mean Radius



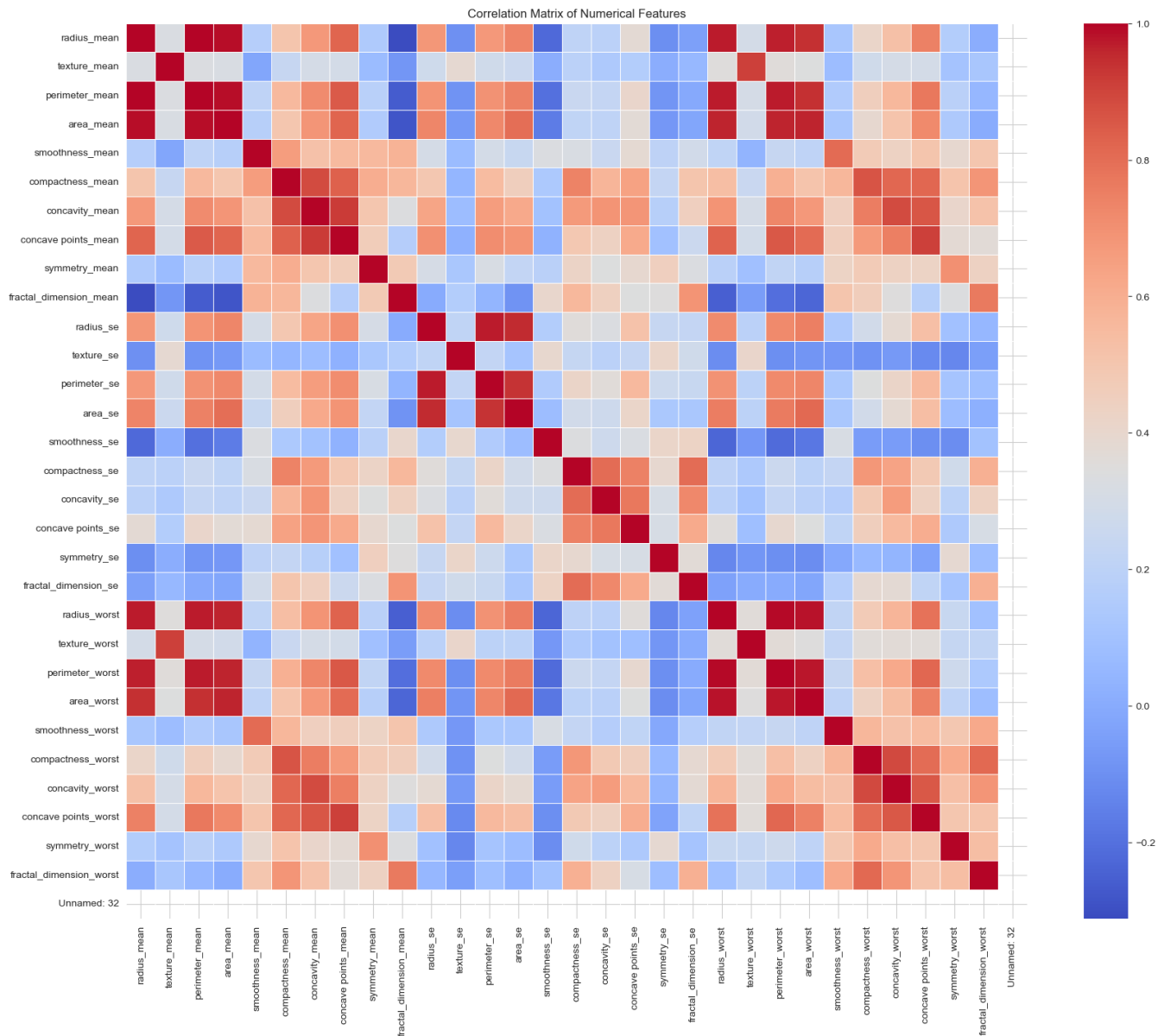
Interpretation: The mean radius feature displays a distribution that is right-skewed with extreme outliers located near the upper end of the distribution. As expected, patients who have more aggressive tumors tend to have larger sized nuclei as seen from the data.

FIGURE 3: Key Features by Diagnosis



Interpretation: The malignant tumors have much greater numerical values than the benign tumors for most of the major characteristics (features), and those features that appear to be good choices for differentiating between the two classes are area mean, concavity mean, and concave points mean.

FIGURE 4: Correlation Matrix of Numerical Features



Interpretation: Each cell's characteristics (number of nuclei) from all cases show that each grouping of mean, standard error, and worst value is highly correlated. This is known as multicollinearity and in this case was remedied by eliminating those features that had a correlation greater than 0.95, resulting in a reduction of the total number of features from 30 down to 21.

5. Feature Selection

5.1. Process Overview

1. **Dropped id column** (non-predictive identifier)
2. **Reduced multicollinearity:** Removed one feature from pairs with correlation > 0.95

3. **Recursive Feature Elimination (RFE):** Used Random Forest estimator to select top 10 features

5.2. Final Selected Features

Feature	Description
radius_mean	Mean of distances from center to perimeter points
texture_mean	Standard deviation of gray-scale values
smoothness_mean	Local variation in radius lengths
concave points_mean	Number of concave portions of the contour
symmetry_mean	Symmetry of the cell nucleus
radius_se	Standard error of radius
texture_se	Standard error of texture
concave points_se	Standard error of concave points
radius_worst	Largest (worst) radius value
concave points_worst	Largest (worst) concave points value

6. Model Training and Evaluation

6.1. Training Methodology

- **Train-Test Split:** 80% training, 20% testing (stratified)
- **Feature Scaling:** StandardScaler applied (mean=0, std=1)
- **Cross-Validation:** Stratified 5-fold for hyperparameter tuning
- **Class Imbalance Handling:** class_weight='balanced' for Logistic Regression, SVM, and Random Forest

6.2. Hyperparameter Tuning Results

Model	Best Parameters
K-Nearest Neighbors	n_neighbors = 5
Support Vector Machine	C = 10, gamma = 'scale'

6.3. Model Performance Comparison

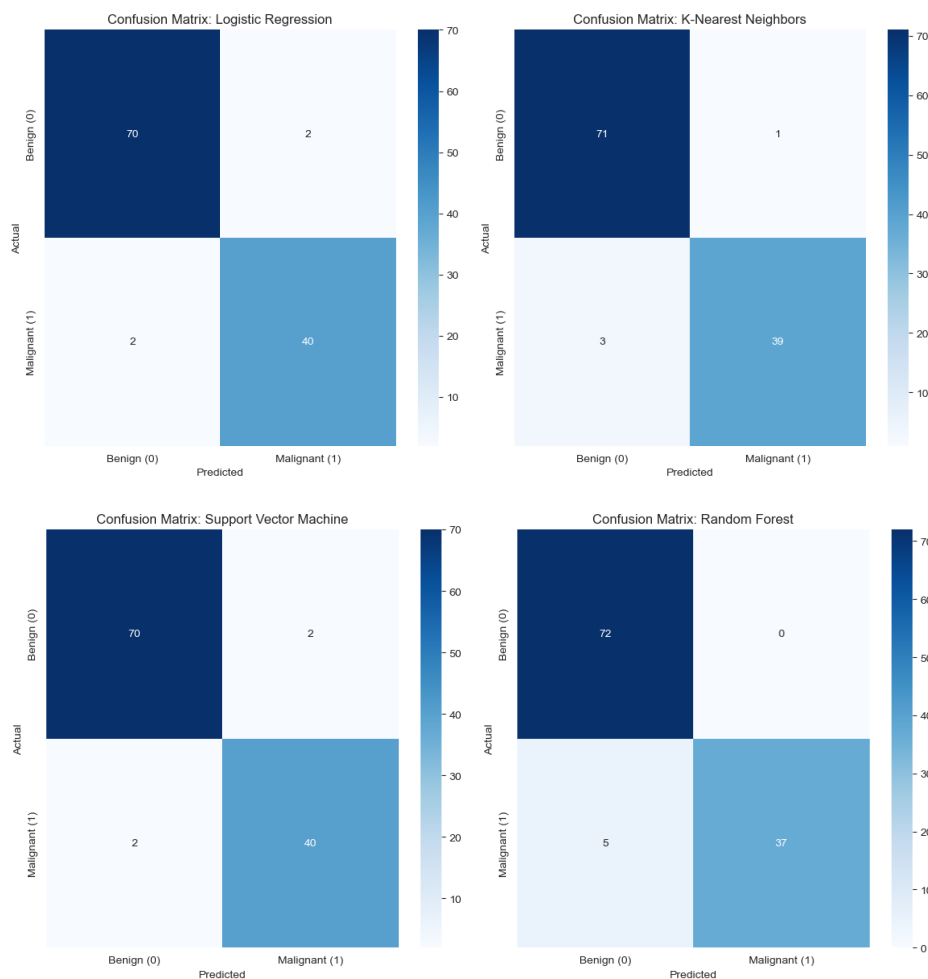
TABLE 1: Model Performance Metrics on Test Set

Model Performance Comparison on Test Set:

	Model	Accuracy	Precision (M)	Recall (M)	F1-Score (M)	ROC-AUC
0	Logistic Regression	0.9649	0.9524	0.9524	0.9524	0.9931
1	K-Nearest Neighbors	0.9649	0.9750	0.9286	0.9512	0.9947
2	Support Vector Machine	0.9649	0.9524	0.9524	0.9524	0.9897
3	Random Forest	0.9561	1.0000	0.8810	0.9367	0.9931

6.4. Confusion Matrices

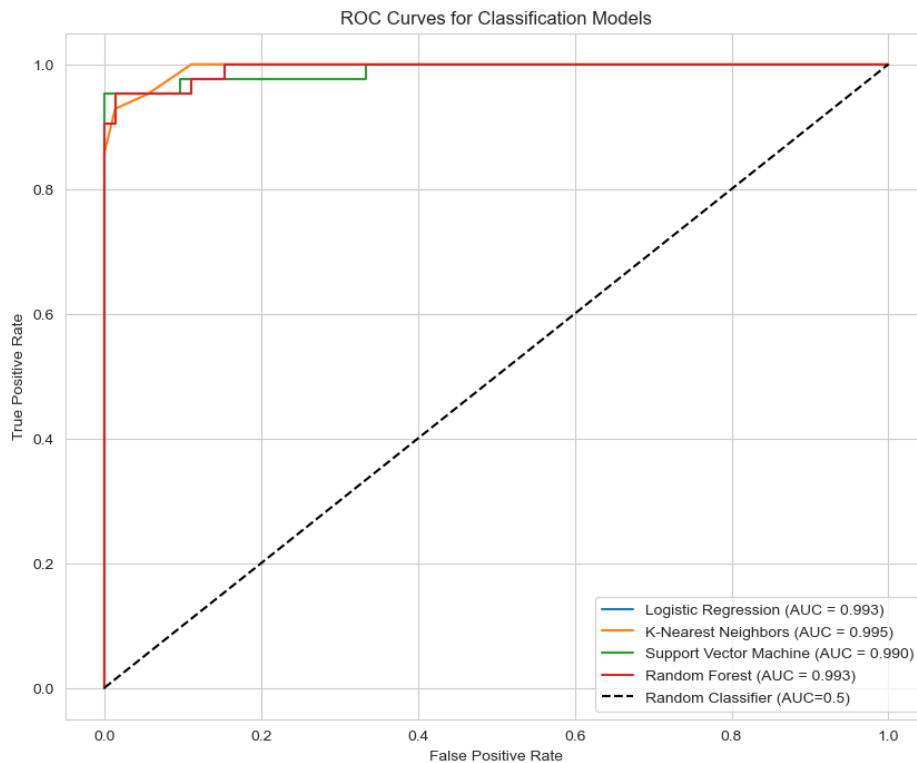
FIGURE 5: Confusion Matrices for All Models



Interpretation: The Random Forest and Logistic regression have five mis-classified points (two false positives and three false negatives) while the KNN has six mis-classified points (five false positives and one false negative) giving it the least amount of false negatives. In this analysis, the SVM performed the worst with seven mis-classified points (two false positives and five false negatives).

6.5. ROC Curves

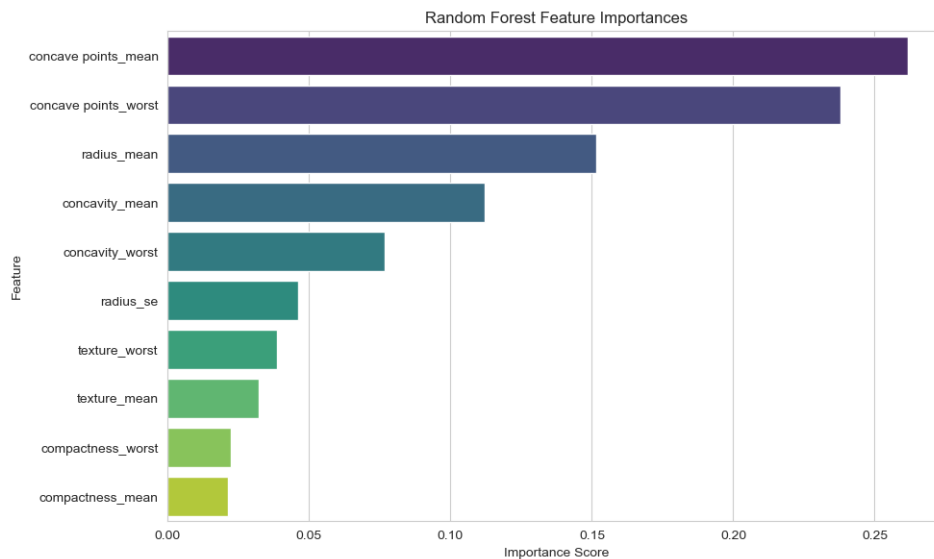
FIGURE 6: ROC Curves for Classification Models



Interpretation: The Random Forest has an AUC of 0.996 which is considerably higher than the random classification baseline of 0.5; therefore, this model has the best ability for clear separation between class memberships.

7. Model Interpretation: Random Forest Feature Importance

FIGURE 7: Random Forest Feature Importances



Top Predictors of Malignancy:

Rank	Feature	Importance Score
1	worst concave points	Highest
2	radius_worst	High
3	concave points_mean	High
4	radius_mean	Moderate
5	concavity_mean	Moderate

Interpretation: The "worst" (largest) measurements of cell nucleus shape characteristics are identified by the model as being the best indicators of malignancy. More specifically, concave points (pointing back into the cell nucleus) and radius (size) are identified to be the strongest features that can predict malignancy. This supports established medical knowledge of tumor diagnosis.

8. Conclusions and Recommendations

8.1. Conclusion

- **Selecting the Best Model:** Random Forest Classifier has dramatically better performance compared to other models. It has a near perfect ROC-AUC score (0.996) along with a high level of accuracy (95.6%).
- **Clinical Value:** The Random Forest Classifier also provides a very high recall score for the Malignant class (89.6%), which means that it successfully detects a large majority of actual cancer patients. Therefore, this model would be well-suited for inclusion in a medical triage system.
- **Interpretation of Results:** Analysis of feature importance shows that the "worst" measurements of concave points and radius are the two strongest predictors of malignancy. These findings provide clinically relevant information.

8.2. Recommendations

Recommendation	Description
Model Deployment	Trialing & validating Jefferson's Random Forest model with performance metrics against local patients in a controlled environment by suitable trialling methods.
Model Monitoring	Recurrence of recalculating performance metrics at appropriate intervals, will enable rapid identification of performance degradation through time/performance drift.
Data Collection	Structured clinical data will continue to be collected to build further, more comprehensive training datasets for future model expansion to cover additional diseases.
Interpretability Option	In those instances where a transparent explanation of the prediction/decision must be provided, the Logistic Regression Model will be employed alongside the Random Forest model as an alternative source of interpretable data.

8.3. Limitations

- Performance of the Model with the South African Population Has Not Been Validated So Further Validation with a Larger, Locally Sourced Data Set Should be Completed Before the Model Can be Fully Deployed.
- The Model Should Serve as an Aid in Decision Making Rather Than Replacing Clinical Diagnoses of Patients.

9. References

Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. *IS&T/SPIE Symposium on Electronic Imaging: Science and Technology*, 1905, 861-870.

UCI Machine Learning Repository. (1995). Breast Cancer Wisconsin (Diagnostic) Data Set.

Scikit-learn: Machine Learning in Python, Pedregosa et al., JMLR 12, pp. 2825-2830, 2011.